

ABOUT

ENGINEER WITH A FOUNDATION IN AUTOMATION, MACHINE SOFTWARE, AND INDUSTRIAL SYSTEMS. I HAVE BUILT MY CAREER ACROSS MANUFACTURING, ROBOTICS, AND EMBEDDED TECHNOLOGIES. I VALUE CLARITY, AND CONTINUOUS IMPROVEMENT. I AM SEEKING A ROLE THAT OFFERS STRONGER OWNERSHIP AND THE OPPORTUNITY TO CONTRIBUTE TO IMPACTFUL ENGINEERING WORK.

EDUCATION

JAN 2016-17	MS, ELECTRICAL & COMPUTER ENGINEERING, (3.41/4) 🏛️ NYU TANDON SCHOOL OF ENGINEERING
JUN 2010-14	BE, INSTRUMENTATION & CONTROLS, (7.04/10) SVIT, GUJARAT TECHNOLOGICAL UNIVERSITY

SKILLS

COMPUTER PROGRAMMING:	JAVA(PRIMARY), C/C++/CS (SECONDARY), GIT, PYTHON AUTOMATION, WEB(HTML, CSS, JS), DESKTOP(JAVAFX, AWT)
AUTOMATION:	MICRO-CONTROLLERS & SINGLE BOARD COMPUTERS, PLCs(LADDER-LOGIC, FBD), HMI/SCADA
INDUSTRIES SERVED:	MACHINERIES, MANUFACTURING, CONTROL PANELS, SOLAR ENERGY, EDUCATION

WORK EXPERIENCE

🔗 SOFTWARE ENGINEER III - AUTOMATION, UNIVERSAL INSTRUMENTS, NY Sep 22 - present

- Develop and maintain software for the **Fuzion** platform.
- Implement **side-barcode scanning** using the Keyence SR-X300 for improved part traceability.
- Redesign the **Wafer AMS** (Accuracy Management System) to improve placement accuracy.
- Build **Selective Placement** with part buffering, enabling smarter pick/hold/place decisions.
- Cognex **Quick-build** integration in Fuzion platform
- Implemented **AVP** (Advanced Vision Processing) to allow user-defined fiducial finds.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree

AUTOMATION ENGINEER , NOR-CAL CONTROLS, CA Jan 22 - Jul 22

- Developed automation software for utility-scale solar power plants
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

SOFTWARE ENGINEER - MACHINE AUTOMATION, MINI-CIRCUITS, NY Feb 18 - Nov 21

- Developed automation software for pick-and-place rotary machines with computer vision inspection.
- Implemented the new shopping cart system for the online store <https://www.minicircuits.com/>
- Developed enterprise application software for internal engineering teams.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript

DEVELOPER INTERN, REVMAX FLEET OPTIMIZATION, NY Sep - Dec 17

- Developed firmware for the SidewalkIQ sensor using Doppler radar to monitor sidewalk activity.
- Deployed sensors at multiple NYC locations and validated field performance.
- Conducted R&D to improve data consistency and optimize storage and transfer to backend servers.
- Tech stack : · Arduino-C ·

INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT., 🏛️ NYU May - Aug 17

- Mentored to Robotics Club Programming Team for Robotics competitions (First Robotics & Vex Robotics).
- Taught concepts of Computer Programming to Robotics Club Team.

LEAD INSTRUCTOR (CREST PROGRAM), CENTER FOR K12 STEM EDUCATION, 🏛️ NYU Sep 17 - Dec 17

- Training High School Interns with Computer Programming (Arduino C), IoT, Robotics & 3D-Printing activities.
- Led STEM projects including :
 - Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
 - Wireless Robot Controller - Arduino: <https://github.com/sap586/WirelessRobotDay5Activity>

DESIGN ENGINEER, PRIMA AUTOMATION, IN Jun 14 - Jan 16

- Programmed Siemens PLCs using TIA Portal and developed HMI screens for industrial automation projects.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.